Progressive block-size sweeps across circuit families

Five families × two inits on DIV2K-8q ($m = n = 8$, 256×256)

Generated 2026-05-24. Test PSNR @ $\rho = 0.20$ (dB) per curriculum stage. Cells: value = final; — = sweep still running; n/a = stage undefined for that family.

1 The experiment

Each `*_progressive` sweep trains a circuit family under a block-size **curriculum**: at stage k the inner basis is a $2^k \times 2^k$ circuit replicated across the 256×256 image by a `BlockedBasis` wrapper (bare full circuit at $k = 8$). Stages are independent (no warm-start); each trains 1008 steps (generalized preset, 112 epochs, `MSELoss` top-k at $\rho = 0.2$).

This run completes the 2-D matrix of **family** × **initialisation**:

Families	`rich` (complex $U(4)$), `qft`, `tebd`, `entangled_qft`, `mera`
Inits	**identity** — every gate dropped to its manifold identity (H → I_2, controlled-phase → phase 0), so the circuit starts as the identity operator. **random** — `rich`: Haar $U(2)/U(4)$; `qft`: Haar $U(2)$ on Hadamards + random controlled-phase angles; `tebd`/`entangled_qft`/`mera`: native seeded random gates. Per-stage seed = seed $+k$.
Stage range	`rich`/`qft`: $k = 1..8$; `tebd`/`entangled_qft`: $k = 2..8$; `mera`: $k \in \{2, 4, 8\}$ only (m must be a power of 2). Table shows $k = 2..8$.

2 Results

family	init	k=2 (4)	k=3 (8)	k=4 (16)	k=5 (32)	k=6 (64)	k=7 (128)	k=8 (256)
rich	identity	32.948	33.657	33.583	33.402	33.395	33.575	33.568
rich	random	32.938	33.351	33.465	32.88	33.153	32.886	33.446
qft	identity	32.046	32.261	31.97	31.66	31.66	31.66	31.659
qft	random	32.046	32.261	31.66	30.906	30.808	31.293	31.293
tebd	identity	32.046	32.261	31.97	31.66	31.66	31.66	31.659
tebd	random	32.046	32.261	31.66	30.906	30.808	30.906	30.906
ent-qft	identity	32.046	32.261	31.97	31.66	31.66	31.66	31.659
ent-qft	random	32.046	32.261	31.659	30.906	31.294	31.293	31.294
mera	identity	32.046	n/a	31.97	n/a	n/a	n/a	31.659
mera	random	32.046	n/a	31.66	n/a	n/a	n/a	30.906

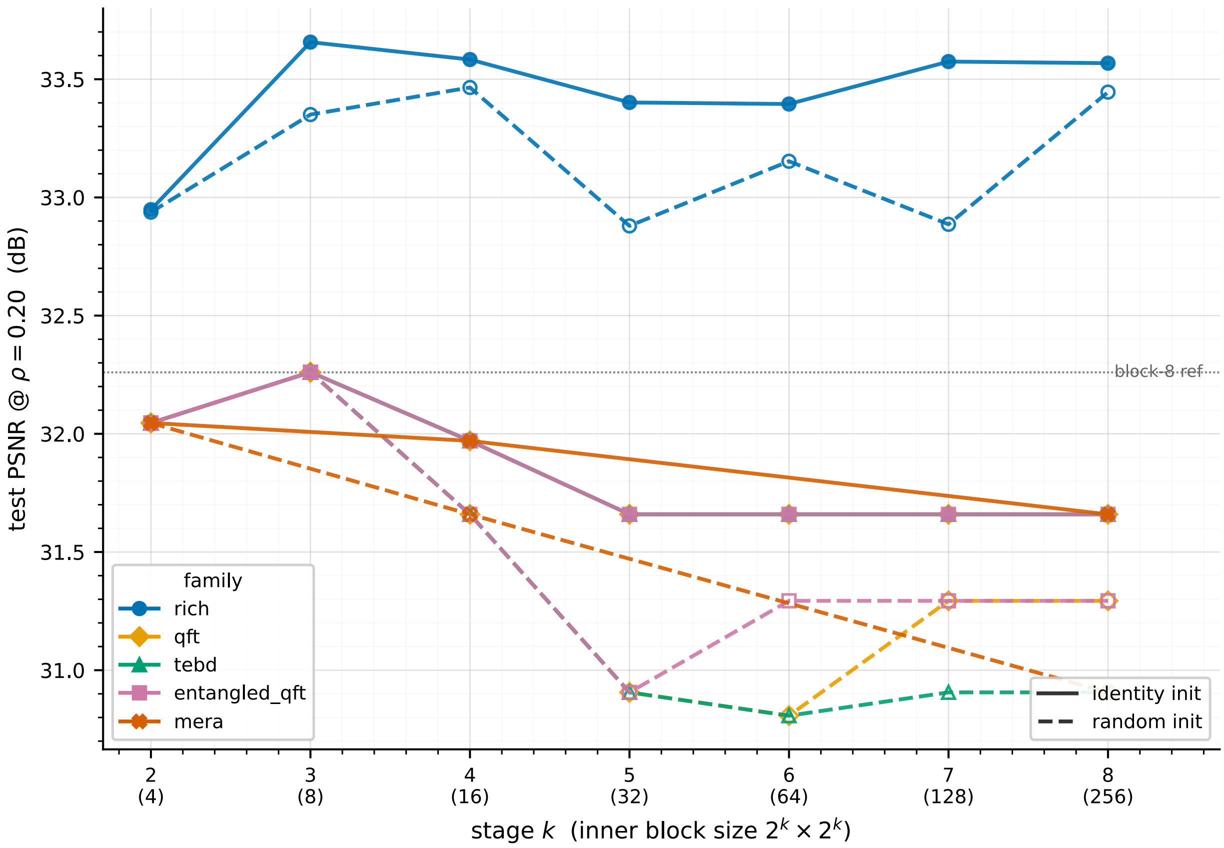

Figure 1: Test PSNR @ $\rho = 0.20$ vs curriculum stage. One colour per family, solid = identity init, dashed = random init (open markers). `mera` appears only at its valid $k \in \{2, 4, 8\}$. **The four solid QFT-family curves (qft, tebd, entangled_qft, mera identity) coincide exactly** and plot on top of one another near 31.7–32 dB; only `rich` (blue) clears the block-8 line.

3 Observations

- **Only `rich` clears the classical block-8 reference.** Its complex $U(4)$ gates reach 33.4–33.7 dB, a full 1.5–2 dB above every other family ($\approx$ 31.7–32). The extra expressivity of the complex two-qubit gates is the one thing in this study that genuinely buys reconstruction quality.
- **Under identity init, the four QFT-derived families are bit-identical at every stage** — `qft`, `tebd`, `entangled_qft`, `mera` all converge to 32.05/32.26/31.97/31.66…/31.66 despite very different gate counts (at $k = 8$: qft 72, tebd 32, entangled_qft 80, mera 44 trainable gates). From an identity start under the top-k MSE objective they all collapse onto the **same** QFT operator; the extra brick-wall / entangling / MERA gates train back to no-ops. Identity init is a strong attractor to the QFT solution.
- **Identity init beats random init for every family.** Randomising the start lands the optimiser in worse, family-specific basins: the QFT-derived families drop $\approx$ 0.5–1 dB at the larger blocks ($k \geq 5$) and stop coinciding, while `rich` (random) shadows `rich` (identity) within $\approx$ 0.5 dB but never beats it. The complex-$U(4)$ freedom adds capacity; randomising the initialisation does not.
- **Small blocks are the great equaliser.** At $k = 2, 3$ every non-`rich` family sits at exactly 32.05/32.26 regardless of init — the 4×4 / 8×8 inner block has a single dominant top-k optimum that everything finds.

Working artefact (report-first; not yet cellified into `results/`). Identity `rich`/`qft` from committed `results/{rich,qft}_progressive`; all other combos from `/tmp/<family>_<init>/`.